

my_predictpca

Create the **my_predictpca** function that we created in the lecture. It takes as input a pca model **mdl** created using the 'my_fitpca' function and **N** feature measurements for **M** samples. It has two outputs, the classification for each sample in the [Mx1] vector **class** that is estimated and the Mahalanobis distance score for each sample in the [Mx1] vector **score** as the second output.

1. Embed your **MahalanobisDistance** function you created in the previous assignment below **my_predictpca** so that you can use
2. Within the **my_predictpca** function, initialize output **class** and **score** to Mx1 zero vectors
3. Loop iteration variable *i* over the M samples, to classify each *i*th sample
 - In a nested loop, loop over each *j*th class to compare the *i*th sample to the *j*th class.
 - Within the loop, find the Mahalanobis distance from the *i*th sample feature vector to the *j*th class pca model
 - The minimum Mahalanobis distance across the class pca models is used to choose our classification for the *i*th sample
 - The classification and the Mahalanobis distance are stored in **class(i)** and **score(i)**

Function ?

Save

Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```

1 function [class,score] = my_predictpca(mdl,data)
2 % Initialize output 'class' and 'score' to Mx1 zero vectors
3 [M, N] = size(data);
4 class = zeros(M,1);
5 score = zeros(M,1);
6 % Loop over the M samples, to classify each 'i'th sample
7 for i=1:M
8     % In a nested loop, loop over each 'j'th class to compare the 'i'th sample to the 'j'th class.
9     for j=1:length(mdl.class)
10        % Within the loop, find the Mahalanobis distance from the 'i'th sample feature vector to the 'j'th
11        mdscore(j) = MahalanobisDistance(mdl.class(j), data(i,:));
12    end
13    % The minimum Mahalanobis distance across the class pca models is used to choose our classification for
14    % The classification and the Mahalanobis distance are stored in class(i) and score(i)
15    [score(i), class(i)] = min(mdscore);
16 end
17 end
18
19 % embed your MahalanobisDistance function you created in the previous assignment here:
20 function [md,b,std_per_mode] = MahalanobisDistance(pcamdl,v)
21 % compute b, our PCA coordinates for v
22 b = pcamdl.eigvectors*(v - pcamdl.mu)';
23 % compute D, the standard deviation distance from the mean for v for each mode of variation
24 std_per_mode = abs(b) ./ sqrt(pcamdl.eigvals);
25 % compute md, the Mahalanobis distance
26 md = sqrt(sum(std_per_mode.^2));
27 end

```

Code to call your function ?

Reset

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17 load fisheriris
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[▶ Run Function](#)

Assessment: All Tests Passed

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✓ Are class and score vectors correct with different inputs?

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